

Lab 3: Component Modelling & Architectural Pattern Selection

Assigned Problem Statement #45 – Multi-Cloud Asset Cost Optimization Portal

The portal is a FinOps analytics system that ingests AWS, Azure, and GCP billing CSVs, identifies unattached storage volumes and idle compute instances, and dispatches budget breach alerts. The target stakeholders are the FinOps Analyst and Cloud Administrator.

Architecture Selection: Microservices Architecture

Architectural Choice: We chose Microservices Architecture for the Multi-Cloud Asset Cost Optimization Portal. The architecture separates ingestion, analysis, storage, dashboard access, cloud-provider integration, and alerting into independently deployable components.

Reason 1 – Multi-cloud integration: AWS, Azure, and GCP can be handled through separate cloud-provider adapters. A change in one provider's API or billing format can be isolated to its adapter without requiring the entire portal to change.

Reason 2 – Independent scaling and fault isolation: Billing ingestion and asset analysis may process large billing files and resource data. These services can be scaled independently, while a failure in one service is less likely to stop the complete portal.

Security Advantage: Services can enforce authentication and authorization at their APIs and restrict access between components. Cloud-provider credentials can be isolated within the provider-adapter layer instead of being exposed to the dashboard.

Performance Benefit: Resource-intensive ingestion and analysis can run independently from dashboard requests. This allows processing workloads to scale separately and helps keep dashboard responses responsive.

Identified Components

Component	Responsibility
FinOps Dashboard	Provides the user interface for FinOps Analyst and Cloud Administrator.
Billing Ingestion Service	Ingests AWS, Azure, and GCP billing CSV files and normalizes billing data.
Asset Analyzer	Detects idle compute instances, unattached storage, and estimates potential savings.
Cost & Asset Data Store	Stores normalized billing, asset, utilization, and analysis results.
Cloud Provider Adapters	Communicates with AWS, Azure, and GCP APIs to obtain resource information.
Alert Management / Dispatcher	Evaluates budget breach conditions and dispatches alerts.

Key Interfaces

1. Dashboard API – dashboard requests analysis results and cost information.
2. Billing Data Interface – ingestion service writes normalized billing data to the data store.
3. Cloud Provider API – provider adapters communicate with AWS/Azure/GCP services.
4. Analysis Results Interface – Asset Analyzer reads/writes asset-analysis results.
5. Alert Interface – alert management uses analysis and budget information to dispatch notifications.